

BACHU, China

WMO: 517160

Lat: 39.7975N

Lon: 78.5711E

Elev: 3665

StdP: 12.85

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.5	9.0	-2.7	5.4	18.3	0.6	6.5	19.5	11.4	29.4	9.7	28.7	1.2	160	0.414

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.9	96.5	65.4	94.1	64.4	91.8	63.7	69.0	88.5	67.7	87.1	66.5	85.8	3.9	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	98.9	77.0	61.5	93.5	75.5	60.0	88.5	74.9	35.6	88.2	34.4	87.0	33.3	86.1	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	10.5	8.9	DB	3.1	101.4	5.4	2.1	-0.8	102.9	-3.9	104.2	-7.0	105.4	-10.9	106.9	(4)
(5)			WB	2.5	71.6	5.1	2.3	-1.2	73.2	-4.2	74.5	-7.1	75.8	-10.8	77.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.5	21.8	32.7	49.1	62.7	70.9	77.1	79.6	77.6	69.4	56.7	40.7	26.1		
	DBStd	20.85	5.65	6.87	7.45	6.85	5.79	4.82	4.85	4.90	4.72	6.19	6.64	4.97		
	HDD50	2505	873	484	109	5	0	0	0	0	0	11	284	739		
	HDD65	5087	1338	904	492	119	18	0	0	0	0	267	730	1204		
	CDD50	4507	0	0	83	385	648	814	918	856	581	218	4	0		
	CDD65	1613	0	0	0	49	201	365	453	391	146	8	0	0		
	CDH74	17429	0	0	21	652	2157	4019	5107	4022	1324	127	0	0		
	CDH80	7786	0	0	1	188	829	1911	2575	1859	411	12	0	0		
	WSAvg	3.3	2.4	3.0	4.1	4.3	4.2	4.0	3.7	3.4	3.1	2.6	2.3	2.2		
	PrecAvg	2.0	0.0	0.0	0.0	0.1	0.3	0.4	0.5	0.4	0.2	0.1	0.0	0.0		
(15) Precipitation	PrecMax	4.9	0.3	0.6	0.9	0.9	1.0	2.0	1.6	1.9	0.8	0.8	0.2	0.2		
	PrecMin	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	1.2	0.0	0.1	0.2	0.2	0.4	0.5	0.4	0.5	0.2	0.2	0.0	0.0		
	PrecStd	1.2	0.0	0.1	0.2	0.2	0.4	0.5	0.4	0.5	0.2	0.2	0.0	0.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.8	55.9	76.8	89.6	93.6	97.9	101.0	98.1	91.3	81.6	66.2	45.6		
		MCWB	31.5	39.4	50.0	56.3	60.2	64.1	67.0	67.2	63.4	56.0	46.2	35.2		
	2%	DB	38.2	51.5	71.6	85.1	90.7	95.1	97.6	94.7	87.6	77.6	61.7	41.3		
		MCWB	29.2	37.7	47.6	54.7	58.8	62.7	66.4	66.4	61.5	55.1	43.9	32.3		
	5%	DB	34.8	48.5	67.8	81.3	88.0	92.7	94.7	92.2	84.8	73.9	57.9	38.2		
		MCWB	27.1	36.1	46.0	53.1	57.9	62.2	65.6	65.8	60.9	53.6	42.5	30.3		
	10%	DB	31.8	44.8	63.6	77.3	85.0	90.1	91.9	89.6	81.7	70.2	53.4	35.1		
		MCWB	25.4	33.9	44.3	51.3	56.9	61.8	64.9	65.3	59.8	51.4	40.8	28.6		
	0.4%	WB	31.8	41.5	50.5	57.6	63.5	67.4	71.1	71.3	66.6	58.5	47.1	35.6		
		MCDB	41.7	52.6	74.5	86.2	86.3	89.1	89.8	90.0	84.7	75.4	63.5	44.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	29.4	38.4	48.2	55.7	60.9	65.6	69.2	69.2	64.0	56.5	44.7	32.9		
		MCDB	37.7	50.0	69.4	81.6	84.8	87.4	88.5	88.7	81.3	74.5	59.4	40.4		
	5%	WB	27.4	36.4	46.5	54.0	59.4	64.3	68.0	67.8	62.6	54.8	42.9	30.6		
		MCDB	34.3	47.6	66.1	78.0	82.7	85.8	87.8	86.5	78.9	71.5	57.0	37.7		
	10%	WB	25.6	34.4	44.8	52.4	58.2	63.2	66.8	66.6	61.2	52.8	41.1	28.7		
		MCDB	31.5	44.4	62.4	74.6	80.7	84.3	86.6	85.0	77.5	67.6	53.3	34.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.3	20.6	21.2	22.6	22.9	22.7	21.9	21.3	22.9	25.9	23.5	18.4		
		MCDBR	23.5	24.0	26.9	30.0	28.7	26.2	25.8	25.4	28.1	29.1	29.1	22.3		
		MCWBR	16.0	13.4	11.6	10.9	9.8	8.3	8.7	9.3	10.5	12.0	14.4	14.6		
		MCDBR	22.8	23.0	25.6	27.2	25.4	23.0	22.6	22.3	25.3	27.3	28.5	21.5		
	5% WB	MCWBR	15.5	13.1	11.1	10.0	9.2	8.1	8.8	9.5	10.3	11.5	14.2	14.1		
		MCWBR	15.5	13.1	11.1	10.0	9.2	8.1	8.8	9.5	10.3	11.5	14.2	14.1		
(40) Clear-Sky Solar Irradiance	taub		0.352	0.409	0.530	0.594	0.618	0.627	0.584	0.637	0.630	0.515	0.427	0.371		
	taud		2.059	1.934	1.661	1.560	1.572	1.593	1.703	1.594	1.567	1.730	1.889	2.007		
	Ebn at Noon		246	246	226	222	221	220	228	210	198	207	212	225		
	Edh at Noon		37	49	72	84	85	83	74	81	78	59	44	36		
(44) All-Sky Solar Radiation	RadAvg		793	1066	1437	1800	2093	2178	2082	1859	1598	1304	921	719		
	RadStd		98	92	100	77	115	105	113	113	90	52	51	62		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-1.30	-1.19	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+0.50	N/A	N/A	N/A	-1.58	-1.91	N/A	N/A	+185	N/A

Nomenclature: See separate page